Linear Algebra I: Week 7

Class examples: 20 worked problems

Fall 2026

How to use these slides

Each problem is centered on one slide; the following slide gives a step-by-step solution. The problems progress from 2×2 matrices to permutations, subspaces, and 3×3 examples.

Problem 1

Compute A^T for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Solution 1

$A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$. Exchange rows and columns: $(A^T)_{12} = a_{21} = 3$.

Problem 2

Compute $(A+B)^T$ for $A = \begin{pmatrix} 1 & 0 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix}$.

Solution 2

$A + B = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$; hence $(A + B)^T = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$. Check $A^T + B^T$.

Problem 3

Find $A^T A$ for $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

Solution 3

$A^T = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$. Multiply: $A^T A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$; it is symmetric.

Problem 4

Find $(AB)^T$ for $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$.

Solution 4

$AB = \begin{pmatrix} 3 & 3 \\ 2 & 6 \end{pmatrix}$, so $(AB)^T = \begin{pmatrix} 3 & 2 \\ 3 & 6 \end{pmatrix}$. Alternatively $B^T A^T$.

Problem 5

Is $A = \begin{pmatrix} 2 & -3 \\ -3 & 1 \end{pmatrix}$ symmetric?

Solution 5

$A^T = \begin{pmatrix} 2 & -3 \\ -3 & 1 \end{pmatrix} = A$. Yes: corresponding off-diagonal entries agree.

Problem 6

Decompose $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ into symmetric and skew parts.

Solution 6

$S = (A + A^T)/2 = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}$, $K = (A - A^T)/2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; $S + K = A$.

Problem 7

Find PA where $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Solution 7

$PA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$. Left multiplication swaps rows 1 and 2.

Problem 8

Find AP for the same A and P.

Solution 8

$AP = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}$. Right multiplication swaps columns 1 and 2; $AP \neq PA$.

Problem 9

Find P^{-1} for $P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$.

Solution 9

$P^{-1} = P^T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$. Verify $P^T P = I$.

Problem 10

Find P exchanging rows 1 and 3 of a 3×3 matrix.

Solution 10

Apply the swap to I_3: $P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$. For any A, PA swaps its first and third rows.

Problem 11

Find P, L, U for $A = \begin{pmatrix} 0 & 2 \\ 3 & 4 \end{pmatrix}$.

Solution 11

$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; $PA = \begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix}$. Thus $L = I_2$, $U = PA$, and $PA = LU$.

Problem 12

Find L, U for $A = \begin{pmatrix} 2 & 1 \\ 6 & 5 \end{pmatrix}$ without row swaps.

Solution 12

$m_{21} = 6/2 = 3$. $R_2 \leftarrow R_2 - 3R_1$ gives $U = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, $L = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$.

Problem 13

Is $W = \{(x, y) : y = 3x\}$ a subspace of $\mathbb{R}^2$?

Solution 13

Yes. $W = \{t(1,3) : t \in \mathbb{R}\}$. Zero corresponds to $t = 0$;
$a(s,3s) + b(t,3t) = (as + bt, 3(as + bt)) \in W$.

Problem 14

Is $S = \{(x, y) : y = 3x + 2\}$ a subspace?

Solution 14

No. $(0, 0)$ fails $0 = 3(0) + 2$. The zero-vector test is sufficient to disprove subspace status.

Problem 15

Is $W = \{(x, y, z) : x + y - z = 0\}$ a subspace?

Solution 15

Yes. For $u, v \in W$, $f(u) = f(v) = 0$ where $f(x, y, z) = x + y - z$. By linearity $f(au + bv) = 0$.

Problem 16

Express $(4, 1)$ as a combination of $(1, 1)$ and $(1, -1)$.

Solution 16

$c_1 + c_2 = 4$, $c_1 - c_2 = 1$. Thus $2c_1 = 5$, $c_1 = 5/2$, $c_2 = 3/2$. Check: $(5/2)(1,1) + (3/2)(1,-1) = (4,1)$.

Problem 17

Find the span of $(1, 2, 0)$ and $(2, 4, 0)$.

Solution 17

Second vector is twice the first. Thus $\text{span}\{(1,2,0),(2,4,0)\} = \{t(1,2,0) : t \in \mathbb{R}\}$, a line.

Problem 18

Find Null(A) for $A = \begin{pmatrix} 1 & 2 & -1 \end{pmatrix}$.

Solution 18

$x_1 + 2x_2 - x_3 = 0$ implies $x_1 = -2s + t$, $x_2 = s$, $x_3 = t$. Hence $\text{Null}(A) = \text{span}\{(-2, 1, 0), (1, 0, 1)\}$.

Problem 19

Find $\mathrm{Col}(A)$ for $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{pmatrix}$; is $(3, 6, 0)$ in it?

Solution 19

$\text{Col}(A) = \text{span}\{(1, 2, 0)\}$ because column 2 is twice column 1. Since $(3, 6, 0) = 3(1, 2, 0)$, yes.

Problem 20

For $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ find A^T and $\det(A)$.

Solution 20

$A^T = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{pmatrix}$. As A is upper triangular, $\det A = 1 \cdot 1 \cdot 2 = 2$; transpose has same determinant.